

Integration at the synapse

These notes follow the lecture slide by slide. Each section names the slides it covers, so you can read it before the video, follow it during the video, or use it afterward to check what you remember. The three worked problems from the slides are written out in full here. Silverthorn chapter 9.

COMPETENCIES 24 TO 28 OF WEEK 4

23 SLIDES · 3 WORKED PROBLEMS · 1 MOVING MODEL

HOW TO USE THIS SHEET

Pass one, before the video: read the section, then close the page and put what you can into the box for that competency on your Competency Study Guide. Pass two, after the video: come back in a different color and add what you missed. Try each problem yourself before you read the answer.

The problem for today

SLIDES 1 TO 2 OF THE DECK

A single neuron in your spinal cord receives thousands of synapses. Some make it more likely to fire and some make it less likely, and at every moment the cell has one decision to make: send an action potential down its axon or stay silent. This lecture is about how that decision is made. We start with the two kinds of small potential a synapse can produce, the EPSP and the IPSP. Then we add them up at the trigger zone, with numbers. Last, we look at three ways the rules can be changed: inhibition of a single terminal, synapses that strengthen with use, and electrical synapses that use no transmitter at all.

Here is the problem we solve by the end. A gardener steps on a rusty nail. A week later his jaw is clamped shut, and the slightest noise or touch throws his whole body into painful spasms. He is awake through all of it. His muscles, and the motor neurons that drive them, are structurally normal. A bacterial toxin from the wound has reached his spinal cord.

Look at what the examination finds. He feels touch, sound and pain normally. He is awake and thinking clearly. His motor neurons carry action potentials, and when a motor neuron fires, its muscle contracts forcefully. Every part that excites a muscle is working. The one thing he has lost is the ability to keep muscles relaxed: a small stimulus makes opposing muscles contract together. Before you read further, write down a prediction. Is the toxin exciting his motor neurons, or is it removing something that normally holds them back?

See it move: will it fire?

SLIDE 13 OF THE DECK

This is a working model of the neuron in this lecture. It has four excitatory inputs, E1 to E4, at different distances along its

dendrites, one inhibitory input on a dendrite, I1, and one inhibitory input on the cell body, I2. The tracing on the right is the membrane potential at the trigger zone, with dashed lines at rest, -70 mV, and at threshold, -55 mV. Press the numbered cases in order, or choose which inputs fire yourself. The readouts beside the model give the highest point the trigger zone reached, how far that was from threshold, and whether the cell fired, and the sentence under the model explains why. Keep it open as you read and come back to it after each section.

Moving model: run it on the course site, [drsrennie-stack.github.io/human-physiology-Fa26/biol005-w04-synaptic-integration-notes.html](https://stack.github.io/human-physiology-Fa26/biol005-w04-synaptic-integration-notes.html)

THREE THINGS TO TRY

- 1 Press case 2, temporal summation. Then move the slider for the time between action potentials from 5 ms to 10 ms and to 25 ms. Write down the highest point each time and compare your numbers with worked problem 2. The model runs slightly higher than the hand calculation, because a real EPSP takes a moment to rise.
- 2 Press case 3, spatial summation, and then slide E4 from 400 out to 800 micrometers. The cell stops firing, and nothing changed except where one synapse sits. That is case C of worked problem 3.
- 3 Press case 4, which adds the inhibitory synapse on the cell body. Read the highest point. Then press Remove all inhibition and watch the same four EPSPs reach threshold. That is the patient in this lecture. Last, turn all the excitatory inputs off, leave I2 on, and move the chloride slider from 10 mM down to 5 mM to see the same synapse change from a small depolarization to a hyperpolarization.

Two kinds of postsynaptic potential

SLIDES 3 TO 5 OF THE DECK

The last lecture ended with one synapse: transmitter binds a receptor, a channel opens, and the postsynaptic membrane potential changes a little. That small change is a graded potential. From the graded potential lecture you know two things about it: it gets smaller as it spreads, and graded potentials can add. You also know that a neuron fires only if the membrane at its trigger zone, where the axon leaves the cell body, reaches threshold, about -55 mV. Synaptic integration is the name for the whole process in between: the neuron adds all of its excitatory and inhibitory inputs and gives one answer.

An excitatory postsynaptic potential, or EPSP, is a small depolarization. At most fast excitatory synapses in the brain, glutamate opens the AMPA receptor, which is a channel for small positive ions. Both Na^+ and K^+ can pass. At -70 mV the force on Na^+ is 131 mV inward, because E_{Na} is $+61$ mV, and the force on K^+ is only 25 mV outward, because E_{K} is -95 mV. So far more Na^+ enters than K^+ leaves, net positive charge comes in, and the membrane depolarizes. If the channel stayed open, the

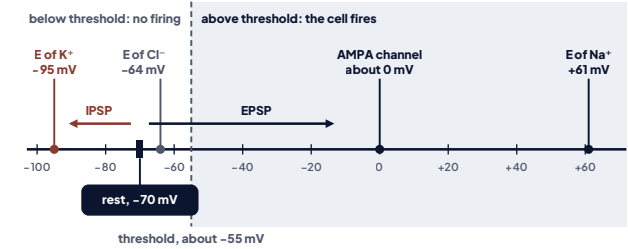
membrane would settle near 0 mV, which is well above threshold. That is what makes this synapse excitatory.

An inhibitory postsynaptic potential, or IPSP, makes the cell less likely to fire. There are two common kinds. One opens K^+ channels. K^+ leaves, and the membrane hyperpolarizes toward E_K , -95 mV. The GABA-B receptor does this through a G protein, so it is a metabotropic receptor from the last lecture, and its IPSP is slow. The other kind opens Cl^- channels. The GABA-A receptor and the glycine receptor are both Cl^- channels that open directly when the transmitter binds, so they are ionotropic and fast. Note that the transmitter does not decide whether a synapse excites or inhibits. The receptor and its channel decide. Acetylcholine excites skeletal muscle through one receptor and slows the heart through another.

Now the part that needs care. Use the rule from the resting potential lecture: opening channels for an ion moves the membrane toward that ion's equilibrium potential. With our concentrations, 110 mM chloride outside and 10 mM inside, E_{Cl} is -64 mV, which is a little above the resting potential of -70 mV. So opening Cl^- channels in our cell moves the membrane only slightly, and it moves it up, toward -64 mV. It is still inhibition, for two reasons. First, the open channels hold the membrane near -64 mV, which is 9 mV below threshold, and the membrane cannot be carried past -64 mV by chloride. Second, when an EPSP lifts the membrane above -64 mV, Cl^- enters through the open channels and cancels part of the positive charge that the EPSP brought in, so the EPSP is smaller than it would have been. This second effect is called shunting inhibition, because the open channels give the excitatory current a path out of the cell.

Cells and books vary on this point, and you should know why. Many adult neurons have a transporter that keeps Cl^- inside nearer 5 to 7 mM. With 110 mM outside, the Nernst equation then gives an E_{Cl} of about -83 to -74 mV, which is below rest, and opening Cl^- channels lets Cl^- in and hyperpolarizes the cell. Silverthorn describes the IPSP that way. Both descriptions are correct for the cells they describe. The general rule covers every case: a synapse is excitatory if its channel moves the membrane toward a value above threshold, and inhibitory if that value is below threshold.

Where each kind of channel moves the membrane, compared with threshold



A synapse is excitatory if its channel moves the membrane toward a value above threshold. It is inhibitory if that value is below threshold.

Figure 2. The equilibrium potentials from the first lecture this week, with threshold added.

Synaptic integration

The process by which a neuron adds all of its excitatory and inhibitory inputs and produces one output: fire or do not fire.

EPSP

Excitatory postsynaptic potential. A small, graded depolarization, usually from a cation channel such as the AMPA receptor letting Na^+ in.

IPSP

Inhibitory postsynaptic potential. A graded potential that makes firing less likely, from opening K^+ channels (hyperpolarizes toward -95 mV) or Cl^- channels (holds the membrane near E_{Cl}).

Shunting inhibition

Inhibition by open channels that let excitatory current leave the cell, so EPSPs are smaller even though the membrane potential itself hardly moved.

GABA-A and glycine receptors

Ionotropic receptors that are Cl^- channels. GABA is the main fast inhibitory transmitter in the brain and glycine in the spinal cord.

GABA-B receptor

A metabotropic receptor that opens K^+ channels through a G protein and gives a slow IPSP.

Working backwards from a recording

SLIDE 6 OF THE DECK

This problem is the reverse of the last section. You are given the recording and asked for the channel. It is worth doing carefully, because it is exactly what you will do in lab with the synapse simulation.

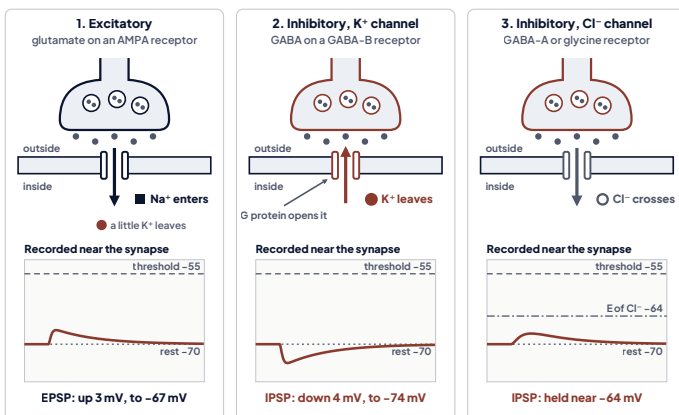


Figure 1. Three synapses and the potential each one produces. Sizes are illustrative. One synapse alone usually gives less than 1 mV.

WORKED PROBLEM 1

An electrode in a neuron records three traces on different trials. Each starts at -70 mV. The first rises 3 mV, the second falls 4 mV, and the third stays flat. For each, name the channel that opened and the ion that moved. Give the flat trace two different explanations, and say what second recording would tell them apart.

You need $E_K = -95$ mV, $E_{Cl} = -64$ mV with our numbers and lower in many neurons, the AMPA channel at about 0 mV, and $E_{Na} = +61$ mV.

THE WORK

- 1 The trace that rises 3 mV.** The membrane moved toward a value above -70 mV, and the channel that does this at a fast excitatory synapse is a cation channel such as the AMPA receptor. It passes Na^+ and K^+ . At -70 mV the force on Na^+ is 131 mV inward and the force on K^+ is only 25 mV outward, so far more Na^+ enters than K^+ leaves. Net positive charge enters, and the trace rises. This is an EPSP.
- 2 The trace that falls 4 mV.** The membrane moved toward a value below -70 mV. With our numbers only E_K , at -95 mV, is below rest, so a K^+ channel opened and K^+ left the cell. A GABA-B receptor does this through a G protein. In a neuron that keeps less chloride inside, a Cl^- channel could give the same trace, with Cl^- entering. Either way it is an IPSP.
- 3 The flat trace, explanation one.** Nothing happened at the synapse. The presynaptic neuron did not fire, or it fired and released no transmitter, so no channel opened.
- 4 The flat trace, explanation two.** A Cl^- channel opened in a cell whose chloride equilibrium potential is equal to its resting potential. There is no net force on Cl^- at that voltage, so no net charge crosses and the voltage does not move, even though the channels are open.
- 5 The second recording that separates them.** Fire a nearby excitatory input at the same moment and compare the EPSP with the EPSP you get from that input alone. If the quiet synapse opened no channels, the EPSP is its usual size. If it opened Cl^- channels, the EPSP is smaller, because as soon as the membrane rises above E_{Cl} , Cl^- enters and cancels part of the positive charge. That is how an input can be inhibitory while the membrane potential barely moves.

about 15 mV above rest. If nothing were lost on the way, it would take about 30 of those EPSPs arriving together to fire the cell. This is very different from the neuromuscular junction, where one action potential in the motor neuron always fires the muscle fiber. In the central nervous system no single synapse decides anything. A spinal motor neuron carries on the order of 10,000 synapses, and its output depends on many of them at once.

The arithmetic of integration can be written in one line. The potential at the trigger zone equals -70 mV, plus the EPSPs arriving, minus the IPSPs arriving, and the cell fires if that value reaches about -55 mV. Arriving means the size of each potential when it reaches the trigger zone, after it has shrunk with distance and with time. Adding potentials as plain numbers is a simplification that works well for small potentials. Real neurons add a little less than the arithmetic says, because each EPSP reduces the force that drives the next one.

There are two ways to add. In temporal summation, one presynaptic neuron fires several times in quick succession. Each EPSP begins before the one before it has decayed, so each starts from a higher level. In spatial summation, several presynaptic neurons fire at about the same moment and their EPSPs arrive at the trigger zone together. Real neurons use both at once: many inputs, each firing a train of action potentials. An EPSP and an IPSP that arrive together cancel, in whole or in part, in the same sum.

Temporal summation depends entirely on timing. After its peak, an EPSP falls by half roughly every 10 ms in the teaching values used here, because charge leaks back out through the membrane. If the next action potential arrives after 25 ms, less than a fifth of the first EPSP is left and almost nothing is added. If it arrives after 5 ms, about 71 percent is left. This is how the frequency code from the lecture on action potentials is read by the next cell. A faster train of action potentials in one axon produces a larger depolarization at the trigger zone of the cell it synapses on.

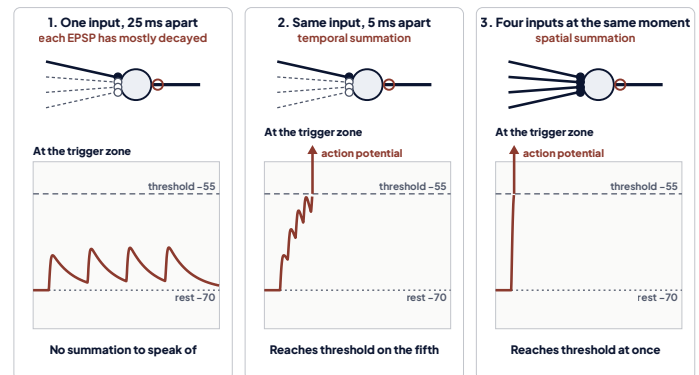


Figure 3. Tracings calculated with the same values as the moving model. Each EPSP is about 5 mV at the trigger zone.

One EPSP is small, so EPSPs have to add

SLIDES 7 TO 9 OF THE DECK

A single excitatory synapse on a neuron in the central nervous system typically produces an EPSP of roughly 0.5 mV at the cell body, and the value varies from synapse to synapse. Threshold is

WORKED PROBLEM 2

One excitatory input gives an EPSP of 5.4 mV at the trigger zone. After its peak, an EPSP falls by half about every 10 ms. The neuron rests at -70 mV and its threshold is -55 mV. Does the cell fire if this one input fires every 25 ms? Every 10 ms? Every 5 ms?

The peak after each new EPSP is 5.4 mV plus whatever is left of the earlier ones. What is left is one half after 10 ms, about 0.71 after 5 ms and about 0.18 after 25 ms.

THE WORK

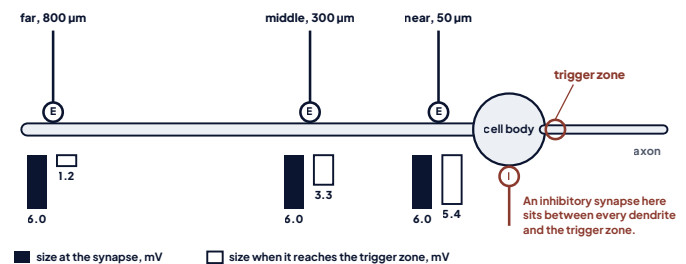
- 1 How far is threshold?** From -70 mV to -55 mV is 15 mV. One EPSP of 5.4 mV reaches -64.6 mV, which is 9.6 mV short.
- 2 Every 25 ms.** After 25 ms an EPSP has been halved two and a half times, so less than one fifth of it is left, about 1 mV. The second EPSP peaks near 6.4 mV above rest, and the later ones go no higher than about 6.6 mV. The cell does not fire however long the train continues.
- 3 Every 10 ms.** Exactly half is left each time. The peaks are 5.4, then $5.4 + 2.7 = 8.1$, then $5.4 + 4.05 = 9.5$, then 10.1 mV, and they level off below $5.4 \times 2 = 10.8$ mV. That is still 4 mV short of threshold.
- 4 Every 5 ms.** After 5 ms about 71 percent is left, because $0.71 \times 0.71 = 0.5$. The peaks are 5.4, 9.2, 11.9, 13.8 and then 15.2 mV. The fifth EPSP carries the trigger zone past 15 mV, and the cell fires.
- 5 What has to be true about timing.** The next EPSP has to arrive before the last one has decayed. The membrane potential takes roughly 10 ms to fall by half, so action potentials more than a few tens of milliseconds apart do not add, and action potentials a few milliseconds apart do. This is how the frequency code from the last lecture is read by the next cell: a faster train in one axon produces a larger depolarization at the trigger zone.

dendrite therefore has less effect at the trigger zone than the same synapse close to the cell body. In this lecture we use one teaching rule for that loss: the fraction that arrives is e raised to the power of minus the distance divided by 500 micrometers. With that rule, a 6 mV EPSP arrives as 5.4 mV from 50 micrometers, 3.3 mV from 300 micrometers and 1.2 mV from 800 micrometers. Real dendrites differ from one neuron to the next, and some have voltage gated channels that boost distant inputs, so treat the rule as an illustration of the direction of the effect.

Location matters for inhibition too. Inhibitory synapses are often found on the cell body and on the first part of the axon. Every EPSP from every dendrite has to pass that point on its way to the trigger zone, so one well placed inhibitory synapse can cancel the combined effect of many excitatory ones.

Two wiring patterns appear all through the nervous system. In divergence, the axon of one neuron branches and synapses on many target cells, so one signal reaches many places. In convergence, many presynaptic neurons synapse on one target cell, which integrates all of them. A spinal motor neuron is a good example of convergence: it receives sensory input, input from the brain and input from local inhibitory interneurons, and its one output decides whether a muscle contracts. The largest example in the body is the Purkinje cell of the cerebellum, which receives on the order of 100,000 to 200,000 synapses. That number is approximate.

The same 6 mV EPSP, started at three different distances from the cell body



Sizes use the teaching rule in this lecture: a graded potential shrinks by the factor $e^{-(\text{distance} / 500 \mu\text{m})}$.

Figure 4. Three identical synapses at different distances, and one inhibitory synapse on the cell body.

FOUR EXCITATORY INPUTS, EACH 6 MV AT ITS OWN SYNAPSE, AND ONE INHIBITORY INPUT

INPUT	DISTANCE FROM THE CELL BODY	FRACTION THAT ARRIVES	SIZE AT THE TRIGGER ZONE
E1	50 μm	0.90	+5.4 mV
E2	150 μm	0.74	+4.4 mV
E3	350 μm	0.50	+3.0 mV
E4	400 μm	0.45	+2.7 mV
I1 (-5 mV at its synapse)	250 μm	0.61	-3.0 mV

Temporal summation

Adding of graded potentials that arrive one after another from the same input, close enough in time that each begins before the last has decayed.

Spatial summation

Adding of graded potentials that arrive at the trigger zone at the same time from different synapses.

Trigger zone

The region where the axon leaves the cell body, rich in voltage gated Na^+ channels. It is where the sum of all inputs is compared with threshold.

Location, and a full worked problem

SLIDES 10 TO 12 OF THE DECK

A graded potential loses size as it spreads, because charge leaks out through the membrane along the way. A synapse far out on a

WORKED PROBLEM 3

Using the table above, with rest at -70 mV and threshold at -55 mV, decide whether the cell fires in each case. A: E1 alone. B: all four excitatory inputs together. C: the same four, with E4 moved out to 800 micrometers, where the fraction that arrives is 0.20. D: case B plus I1. E: case D, but the inhibitory terminal cannot release its transmitter.

Add the sizes at the trigger zone to -70 mV and compare the result with -55 mV.

THE WORK

- 1 **Case A, E1 alone.** $-70 + 5.4 = -64.6$ mV. That is 9.6 mV short of threshold, so the cell does not fire.
- 2 **Case B, all four excitatory inputs at the same moment.** $5.4 + 4.4 + 3.0 + 2.7 = 15.5$ mV. $-70 + 15.5 = -54.5$ mV. That is above -55 mV, so the cell fires. This is spatial summation.
- 3 **Case C, the same four, with E4 moved out to 800 μm .** E4 now arrives as $6 \times 0.20 = 1.2$ mV. The sum is $5.4 + 4.4 + 3.0 + 1.2 = 14.0$ mV, which gives -56.0 mV. The cell does not fire. Nothing about the synapse changed except where it sits.
- 4 **Case D, all four as in case B, plus I1.** I1 makes -5 mV at $250 \mu\text{m}$ and arrives as $-5 \times 0.61 = -3.0$ mV. The sum is $15.5 - 3.0 = 12.5$ mV, which gives -57.5 mV. The cell does not fire. One inhibitory input has canceled the effect of four excitatory ones.
- 5 **Case E, case D with the inhibitory terminal unable to release.** The -3.0 mV is gone, the sum is back to 15.5 mV, and the cell fires again. Keep this case in mind for the patient.
- 6 **The simplification.** Adding potentials as plain numbers is close to right for small potentials and is how these problems are worked in this course. In a real neuron the sum comes out a little smaller than the arithmetic, and an inhibitory synapse that opens Cl^- channels on the cell body removes more than its own size suggests. The model on slide 13 shows that case.

ONE MORE TO TRY ON YOUR OWN

A neuron has three excitatory inputs worth about 4 mV each at the trigger zone and two inhibitory inputs worth about 5 mV each. Can it ever reach threshold?

All three excitatory inputs fired once, at the same moment, give 12 mV, which is 3 mV short. So single volleys cannot do it, even with no inhibition. Temporal summation can. If all three fire twice, 5 ms apart, the second volley adds 12 mV to the $0.71 \times 12 = 8.5$ mV that is left, which is 20.5 mV and well past threshold. Add one inhibitory input at the time of the second volley and the total falls to 15.5 mV, which still fires. Add both and it falls to 10.5 mV, which does not.

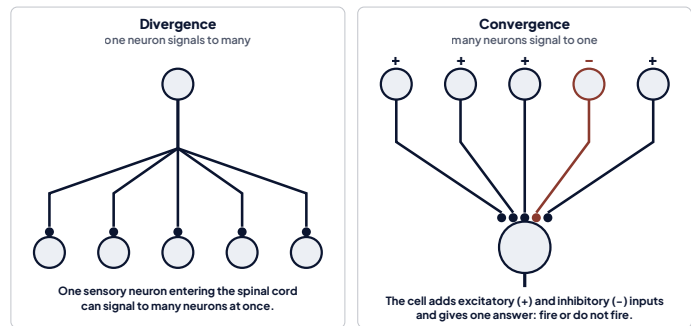


Figure 5. Divergence and convergence.

Divergence

One neuron synapsing on many target cells.

Convergence

Many neurons synapsing on one target cell, which integrates them.

Presynaptic inhibition and facilitation

SLIDES 14 TO 15 OF THE DECK

Everything so far has acted on the receiving cell. There is another place to act. In presynaptic inhibition a modulatory neuron ends on the axon terminal of another neuron, which is called an axoaxonic synapse. Its transmitter reduces the amount of Ca^{2+} that enters that terminal when an action potential arrives. From the last lecture you know that Ca^{2+} entry is what makes vesicles fuse. Less Ca^{2+} entering means fewer vesicles fuse, less transmitter is released, and the EPSP from that one input is smaller. The postsynaptic cell has not changed at all, so it responds to its other inputs exactly as before.

Compare that with postsynaptic inhibition, which is the IPSP from the first section. It acts on the receiving cell, so it reduces the effect of every input to that cell. The advantage of presynaptic control is selectivity: one input can be turned down while the others are left alone. If you needed to cut the effect of input A by half and leave inputs B and C untouched, an inhibitory synapse on the terminal of A would do it. An inhibitory synapse on the cell body would fail the requirement, because it would lower B and C by just as much.

Presynaptic facilitation is the opposite change at a terminal. The modulatory transmitter keeps the terminal depolarized a little longer when an action potential arrives, more Ca^{2+} enters, and more transmitter is released. It was worked out in the sea slug *Aplysia*, where serotonin closes K^+ channels in a sensory terminal.

The standard example of presynaptic inhibition in the human body is pain. Pain fibers make their first synapse in the dorsal horn of the spinal cord. Interneurons there release enkephalins, the body's own opioids, onto the terminals of the pain fibers. Opioid receptors on those terminals reduce Ca^{2+} entry, less transmitter is released, and less of the pain signal reaches the next neuron. Morphine and the other opioid drugs act at the same receptors. Opioids also act at other sites, including on the postsynaptic cell and in the brain, so this is one part of how they work and not the whole of it.

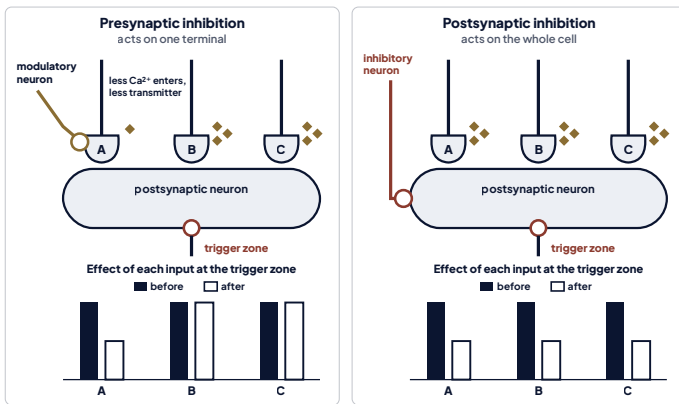


Figure 6. The same three inputs, inhibited in two different places.

Axooxonic synapse

A synapse made by one axon terminal onto another axon terminal.

Presynaptic inhibition

A decrease in transmitter release from one terminal, caused by a modulatory input that reduces Ca^{2+} entry into that terminal. It is selective for that one input.

Presynaptic facilitation

An increase in transmitter release from one terminal, caused by a modulatory input that increases Ca^{2+} entry.

Postsynaptic inhibition

An IPSP in the receiving cell. It reduces the effect of all inputs to that cell.

Synaptic plasticity

SLIDES 16 TO 18 OF THE DECK

Synaptic plasticity means that the strength of a synapse depends on its history. The same presynaptic action potential can produce a larger or a smaller EPSP than it did before. Some of these changes are brief. In facilitation, each EPSP in a short rapid train is larger than the one before, because Ca^{2+} from one action potential has not been cleared from the terminal when the next arrives, so more vesicles fuse. In depression, the EPSPs get smaller during a long rapid train, because the terminal is using vesicles faster than it can refill them. Both changes are in the presynaptic terminal, and both fade within seconds to minutes after the train stops. A memory lasts far longer than that, so there must also be changes that last.

Long term potentiation, or LTP, is an increase in synaptic strength that lasts hours or longer. It was discovered in the hippocampus, a part of the brain needed for forming new memories, and it is best understood at glutamate synapses. Take it one step at a time. Glutamate binds two receptors on the postsynaptic membrane. The AMPA receptor opens at once and produces the ordinary EPSP. The NMDA receptor also binds glutamate, but near the resting potential its channel is plugged by a Mg^{2+} ion from the outside fluid, which the negative interior holds in the pore. So after one action potential, the NMDA receptor has glutamate on it and passes almost nothing.

A rapid burst of input changes that. Many EPSPs add, by the temporal and spatial summation you have already worked through, and the membrane depolarizes strongly. A less negative interior no longer holds the positive Mg^{2+} ion, so the plug leaves

the pore. With glutamate still bound, the NMDA channel is now open, and it lets Ca^{2+} in as well as Na^{+} . Ca^{2+} acts as a second messenger. It activates protein kinases in the postsynaptic cell, and these add more AMPA receptors to the membrane and make the existing ones pass more current. The postsynaptic cell may also send a signal back across the cleft so that the terminal releases more glutamate. Changes that last for days also need new proteins to be made.

The result is that the same single action potential now produces a larger EPSP. Notice why one action potential alone cannot do this. The NMDA receptor opens only when two things are true together: glutamate is bound, which says the presynaptic cell fired, and the membrane is already strongly depolarized, which says the postsynaptic cell was strongly driven at that moment. One EPSP of half a millivolt does not remove the Mg^{2+} .

In the standard experiment, the EPSP is tested with a single action potential once a minute. After one burst of rapid stimulation it jumps, settles at a larger size than before and stays there for as long as the recording lasts. If NMDA receptors are blocked with a drug, ordinary transmission is normal, because AMPA receptors carry it. The burst still causes a brief rise, which is short term plasticity in the terminal, but the lasting increase does not happen. So the AMPA receptor carries the ordinary, moment to moment signal, and the NMDA receptor carries the signal that something should change.

Long term depression, or LTD, is the reverse. Prolonged slow stimulation lets in a small amount of Ca^{2+} , which activates a different set of enzymes. AMPA receptors are removed from the membrane, and the EPSP becomes smaller for a long time. Most neuroscientists think changes like these are part of how learning and memory work, and blocking NMDA receptors in the hippocampus does impair some kinds of learning in animals. How a particular memory is stored is still being worked out, so treat LTP as a well supported mechanism for strengthening a synapse and not as a full explanation of memory.

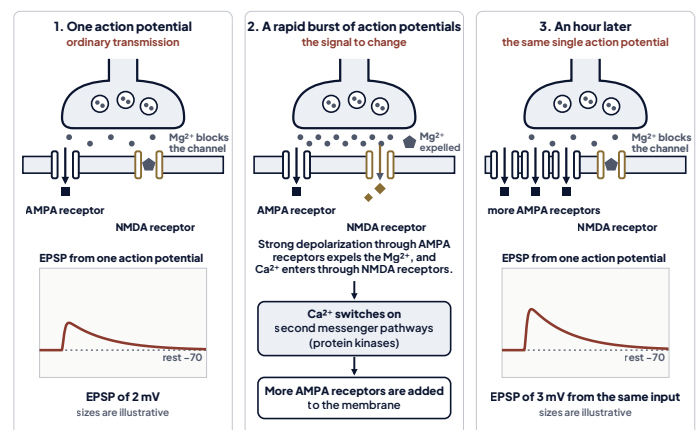


Figure 7. One glutamate synapse before, during and after a rapid burst of activity.

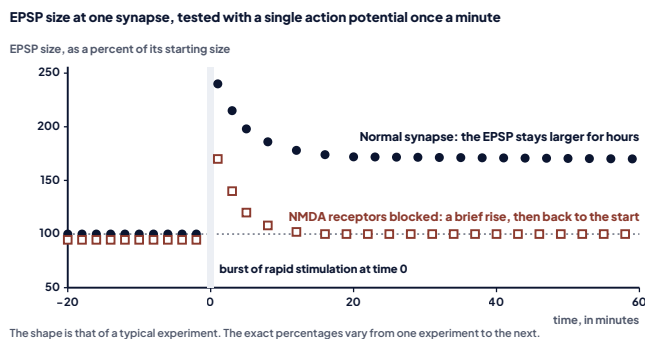


Figure 8. The size of the EPSP before and after one burst of rapid stimulation, with and without working NMDA receptors.

Synaptic plasticity

A change in the strength of a synapse that depends on its earlier activity.

Facilitation and depression

Short term increases and decreases in transmitter release during a train of action potentials, lasting milliseconds to minutes.

AMPA receptor

An ionotropic glutamate receptor that opens at once and produces the ordinary fast EPSP.

NMDA receptor

An ionotropic glutamate receptor whose channel is blocked by Mg^{2+} near rest. It opens only when glutamate is bound and the membrane is strongly depolarized, and it lets Ca^{2+} in.

Long term potentiation (LTP)

A long lasting increase in the size of the EPSP at a synapse after strong, rapid activity. It depends on Ca^{2+} entry through NMDA receptors and on adding AMPA receptors.

Long term depression (LTD)

A long lasting decrease in the size of the EPSP, with AMPA receptors removed from the membrane.

Electrical synapses

SLIDES 19 TO 20 OF THE DECK

Not every synapse uses a transmitter. At an electrical synapse two cells are joined by gap junctions. Each gap junction channel is built from proteins called connexins. Six connexins form a half channel in one membrane, six more form a half channel in the other, and the two line up to make one pore from the cytoplasm of one cell to the cytoplasm of the next. Where they occur, the two membranes lie only about 3.5 nm apart, compared with a cleft of about 20 to 40 nm at a chemical synapse.

Ions carry current straight through the pore, so a depolarization in one cell depolarizes its neighbor with almost no delay. A chemical synapse takes about 0.5 to 1 ms, and sometimes longer, because Ca^{2+} has to enter, vesicles have to fuse, and transmitter has to cross and bind. Current can usually pass through a gap junction in either direction, so a group of cells joined this way tends to fire together. You find electrical synapses in heart muscle, in the smooth muscle of the gut and the uterus, between glial cells, and between some neurons in the brain.

What an electrical synapse gives up is flexibility. The signal keeps its sign, so a depolarization can only pass on as a smaller

depolarization. It cannot be turned into inhibition, it cannot be amplified, and there is no terminal to modulate and no receptor number to change. Gap junctions can open and close, but that is a much coarser control. So the two kinds of synapse suit different jobs. When every muscle cell in the wall of the heart must contract at nearly the same moment, gap junctions are the right choice, because delays at millions of chemical synapses would add up and the contraction would not be coordinated. When a spinal motor neuron must be able to ignore a sensory signal on some occasions and respond to it on others, only a chemical synapse will do. The delay of a chemical synapse is the cost of being able to integrate.

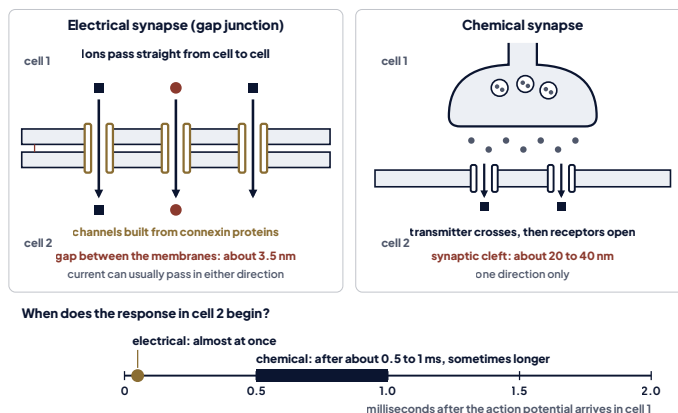


Figure 9. The two kinds of synapse, with the time each one takes.

TWO WAYS FOR ONE CELL TO SIGNAL TO THE NEXT

	CHEMICAL SYNAPSE	ELECTRICAL SYNAPSE
What crosses	Neurotransmitter, across a cleft of about 20 to 40 nm	Ions, through connexin channels across a gap of about 3.5 nm
Delay	About 0.5 to 1 ms, sometimes longer	Almost none
Direction	One way	Usually both ways
Sign of the signal	Excitatory or inhibitory, depending on the receptor	The same sign in both cells
Amplification	Yes	No. The signal can only get smaller
Adjustable	Highly: presynaptic control, receptor number, plasticity	Much less
Best suited for	Integration and decisions	Speed and firing together

Electrical synapse

A junction where current passes directly from one cell to the next through gap junction channels.

Gap junction

A cluster of channels that join the cytoplasm of two cells. Ions and small molecules pass through.

Connexin

The protein that gap junction channels are built from, six in each membrane.

The patient, explained

SLIDES 21 TO 22 OF THE DECK

Start with a normal spinal motor neuron. It receives excitatory input all the time, from sensory neurons and from the brain. Inhibitory interneurons that release glycine or GABA keep it below threshold unless the excitation is strong. Put illustrative numbers on it. An excitatory input worth 18 mV and an inhibitory input worth -8 mV arrive together, and $-70 + 18 - 8 = -60$ mV, so the cell does not fire. Now block release from the inhibitory terminal. $-70 + 18 = -52$ mV, which is above threshold. The same input that did nothing a moment ago now fires the motor neuron.

That is what happened to the gardener. The bacterium *Clostridium tetani* grew in the deep wound and released tetanus toxin. Motor axon terminals took the toxin up, and retrograde axonal transport, from the neuron lecture, carried it back to the cell bodies in the spinal cord and brainstem. That trip is part of why a week went by. In the spinal cord the toxin passed from the motor neuron into the terminals of the inhibitory interneurons that synapse on it. There it cut one of the proteins that a vesicle needs in order to fuse with the membrane, so those terminals could no longer release glycine or GABA.

Nothing was added to the excitatory side. His motor neurons still receive the same EPSPs from every touch and sound, but the inhibitory term is gone from the sum, so ordinary input now reaches threshold. Normally, when one muscle contracts, inhibitory interneurons keep the opposing muscle quiet. Without them, both contract at once and with full force. The strong jaw closing muscles clamp the jaw shut, which gives the disease its old name, lockjaw, and any stimulus sets off a spasm of the whole body.

Treat it as an experiment. The variable that failed is inhibitory transmitter release onto motor neurons, and through it the sum at the trigger zone. Three physiological targets follow from that. The GABA-A receptors on the motor neuron are intact, so a drug that makes those channels open more readily, such as a benzodiazepine, increases whatever inhibition remains and puts a negative term back into the sum. Every sound and touch adds EPSPs, so a quiet, dark room lowers the excitatory side of the same sum. And antibody against the toxin binds whatever toxin has not yet entered neurons. Toxin already inside the terminals cannot be reached, so recovery waits for those terminals to be repaired or replaced, which takes weeks. Vaccination works by having that antibody present before any toxin arrives.

Two comparisons test whether you have the mechanism. Strychnine blocks the glycine receptor on the motor neuron. Glycine is still released, but its receptor cannot open its Cl^- channel, so there is no IPSP, and the result is convulsions set off by small stimuli, very much like tetanus. One agent acts before the cleft and the other after it, and both remove the same term from the sum. Botulinum toxin is a close relative of tetanus toxin and cuts the same family of vesicle proteins, but it stays in the motor neuron terminal at the muscle, where it blocks acetylcholine release. The muscle cannot be excited at all, and the result is limp paralysis. The mechanism is the same, the synapse is different, and the effect is the opposite.

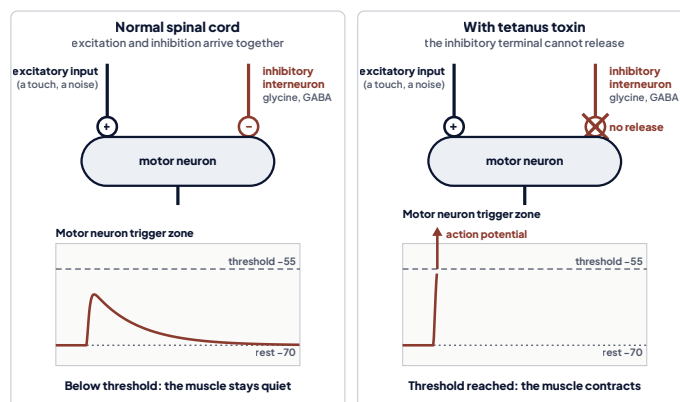


Figure 10. A spinal motor neuron with the same excitatory input in both panels.

What this lecture assumes you already have

If any line here is shaky, fix it before you work the lecture.

CHEMISTRY

Ions and charge: Na^+ , K^+ , Cl^- , Ca^{2+} and Mg^{2+} , and that like charges repel.

Proteins as channels, receptors and enzymes, and what a protein kinase does.

Chemistry review page

FROM EARLIER LECTURES

The resting membrane potential: $E_K -95$ mV, $E_{\text{Na}} +61$ mV, $E_{\text{Cl}} -64$ mV, rest -70 mV, and the rule that opening channels for an ion moves the membrane toward that ion's equilibrium potential.

Neurons and glia: dendrites and cell body as the input zone, the trigger zone, and retrograde axonal transport.

Graded potentials: they shrink with distance and can add, and threshold is about -55 mV.

How action potentials carry information: the frequency code.

The chemical synapse: Ca^{2+} entry and vesicle fusion, ionotropic and metabotropic receptors, glutamate, GABA and glycine.

The only anatomy used is the dorsal horn of the spinal cord, where pain fibers make their first synapse, and the hippocampus and cerebellum, which are named and not described.

MATH

Adding and subtracting signed numbers.

Multiplying by a fraction, for example 6×0.45 .

No logarithms are needed in this lecture. The one Nernst value, E_{Cl} , is given.

Close the page and produce it

These are the Week 4 competencies this lecture covers, straight off the study guide, with the two prompts that go with each one. Competency 29, the synapse simulation, is done in lab. Pick one prompt per competency. Blank paper, nothing open.

Read the full instructions first, including how the first pass before the lecture differs from the second pass after it.

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Excitatory and inhibitory postsynaptic potentials

Distinguish an EPSP from an IPSP by the ion channel opened and the direction of the membrane potential change, and contrast fast responses through ionotropic receptor-channels with slow responses through metabotropic G protein coupled receptors, including how a neuromodulator differs from a neurotransmitter.

Do one of the two. Head your paper with the number.

24-1

Draw a postsynaptic neuron with two synapses on one dendrite, one excitatory and one inhibitory. At each, draw the channel that opened and the ion moving through it with an arrow for direction. Under each synapse draw its own trace starting from about -70 mV and label the peak value. One line: how an inhibitory input can still be inhibitory even when the membrane potential barely moves.

24-2

Draw three recorded traces from the same neuron on shared axes, all starting near -70 mV, one rising 3 mV, one falling 4 mV, and one staying flat. For each trace, work backwards and draw the channel and the ion that would produce it. Give the flat trace two different explanations. One line: which explanation a second recording could rule out, and what that recording would be.

25

Summation and integration

Distinguish temporal from spatial summation and determine whether a stated combination of EPSPs and IPSPs will bring the trigger zone to threshold.

Do one of the two. Head your paper with the number.

25-1

Draw the same neuron twice. On the first, one presynaptic neuron fires four times in quick succession. On the second, four separate presynaptic neurons fire at the same moment. Under each, draw the trace at the trigger zone with a resting line near -70 mV and a dashed threshold line near -55 mV. Label which is temporal and which is spatial. One line: what has to be true about timing for the first one to work at all.

25-2

Draw a neuron with five inputs, three excitatory worth about 4 mV each and two inhibitory worth about 5 mV each. Choose which fire and in what order, then draw the trace at the trigger zone against a threshold line near -55 mV, writing your running total along the trace. Now change one input, its timing or its sign, so the outcome flips. Write which change you made and why it was enough.

26

Presynaptic modulation

Compare presynaptic inhibition and facilitation with postsynaptic modulation and state the advantage of presynaptic control at a single input, which is selectivity.

Do one of the two. Head your paper with the number.

26-1

Draw a neuron with three excitatory inputs converging on it. Add a modulatory neuron that synapses on one input's terminal, not on the cell body. Show what it does to calcium entry and to transmitter release at that terminal only. Draw the postsynaptic response to all three inputs before and after. One line: what postsynaptic inhibition would have done to the other two inputs.

26-2

You need to cut one input's effect by half and leave the other inputs untouched. Draw two attempts side by side, one acting on that input's presynaptic terminal and one acting on the postsynaptic membrane. On each drawing, shade every synapse the attempt affected. Mark the attempt that failed the requirement. One line: name the property that makes presynaptic control the right tool for this job.

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Synaptic plasticity

Explain long term potentiation as a mechanism of learning, tracing how glutamate acts on AMPA and NMDA receptors, how strong depolarization removes the magnesium block, how calcium entry starts second messenger pathways, and how repeated stimulation leads to insertion of new AMPA receptors.

Do one of the two. Head your paper with the number.

27-1

Draw a glutamate synapse with both AMPA and NMDA receptors in the postsynaptic membrane. Show the magnesium sitting in the NMDA channel at rest. In a second panel, draw the same synapse during repeated high frequency stimulation with the magnesium displaced and calcium entering. In a third, draw what the membrane looks like an hour later. One line: why one presynaptic action potential on its own cannot do this.

27-2

Draw a graph of postsynaptic response size against time, with a burst of high frequency stimulation marked on the time axis, and plot the response before, during, and long after the burst. On the same axes, draw a second curve for the identical experiment with NMDA receptors blocked. One line: which receptor carries the ordinary moment to moment signal, and which one carries the signal that something should change.

28

Electrical synapses

Compare an electrical synapse at a gap junction with a chemical synapse by speed and by capacity for modulation.

Do one of the two. Head your paper with the number.

28-1

Draw two synapses side by side in cross section at the same scale, an electrical one at a gap junction and a chemical one. Show what physically crosses at each and make clear which gap is wider. Underneath both, draw one shared time axis and mark when the postsynaptic response begins for each. One line: what the chemical synapse buys with the delay it pays.

28-2

Pick two real jobs, one where many cells must fire as a single unit and one where a signal must be turned up, turned down, or blocked outright. Draw the synapse you would build for each and label it electrical or chemical. On each drawing, circle the feature that made it right for that job. One line: what fails if you swap the two.

Dr. Sharilyn Rennie · BIO 005 Human Physiology · Integration at the synapse. Potential sizes and distances are teaching values. All figures are original drawings for this course.